Disproof of Conjecture 00000005613

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Abstract

A third-order subresultant coefficient minor can vanish while its corresponding principal minor is nonzero. For the monic, squarefree, coprime, strictly interlacing polynomials $f(x) = x^3 - 4x$ and $g(x) = x^2 - 1$, the first subresultant is $-3x$: its constant-coefficient minor is zero, whereas its principal coefficient is -3. The example uses the actual Sylvester submatrix construction and is verified in Lean 4 using only the standard library.

1 The precise clause and standard convention

Conjecture 00000005613[1] asserts that "a minor is zero if and only if the corresponding principal minor is zero" in a discussion of third-order subresultant minors. We disprove this clause using standard subresultant coefficient minors and their canonical leading principal minor.

For degrees p, q and $i < \min(p, q)$, the i-th Sylvester submatrix has rows given by

$$x^{q-i-1}f, \ldots, f, \quad x^{p-i-1}g, \ldots, g,$$

written as coefficient vectors in descending degree. Its dimensions are $(p + q - 2i) \times (p + q - i)$. For a matrix A, let $M_{k,t}(A)$ be the determinant using its first k rows, first $k - 1$ columns, and column $k + t$ (one-based indexing). Thus $M_{k,0}$ is the leading principal minor. The standard polynomial subresultant is

$$\mathrm{Sres}_i(f, g)(x) = \sum_{j=0}^{i} M_{p+q-2i,i-j}(\mathrm{Syl}_i(f, g))x^j;$$

its principal coefficient is $M_{p+q-2i,0}$. These conventions are stated explicitly in Kerber, Definitions 2.3.10–2.3.12[1].

The printed conjecture does not separately define "spectrum," "conversion," or "corresponding." This disproof uses the standard coefficient/principal correspondence above. It makes no claim about an unspecified alternative correspondence or about the other undefined spectral assertions. In particular, the example is not merely an arbitrary matrix with unrelated minors. The target is an *individual* coefficient minor, not the vanishing of the whole subresultant polynomial. In this example the whole first subresultant and its principal coefficient are both nonzero; that different equivalence is not being contradicted.

[1] Official conjecture 00000005613 on GitHub, checked 20 October 2026.

2 The counterexample

Take

$$f(x) = x^3 - 4x = (x+2)x(x-2), \qquad g(x) = x^2 - 1 = (x+1)(x-1).$$

Their roots strictly alternate:

$$-2 < -1 < 0 < 1 < 2.$$

Both polynomials are monic and have distinct real roots. The coefficient-level identity

$$xf(x) + (3 - x^2)g(x) = -3 \tag{1}$$

proves they are coprime over $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$: a common polynomial divisor would divide the nonzero constant -3.

Here $p = 3$, $q = 2$, $i = 1$, so the relevant minors have order $p + q - 2i = 3$. The rows of the first Sylvester submatrix are f, xg, g, in the basis $x^3, x^2, x, 1$:

$$S = \mathrm{Syl}_1(f, g) = \begin{pmatrix} 1 & 0 & -4 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}.$$

This matrix is generated from the polynomial coefficients and the prescribed shifts. Its principal coefficient is

$$M_{3,0}(S) = \det \begin{pmatrix} 1 & 0 & -4 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = -3 \neq 0. \tag{2}$$

The corresponding constant-coefficient minor is

$$M_{3,1}(S) = \det \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix} = 0. \tag{3}$$

The first determinant equals -3 by expansion along its second column. The second is zero because its first two rows agree. Consequently

$$\mathrm{Sres}_1(f, g)(x) = M_{3,0}(S)x + M_{3,1}(S) = -3x.$$

Theorem 1. *The vanishing-equivalence clause of Conjecture 00000005613 is false under the standard subresultant coefficient/principal convention, even for monic squarefree coprime polynomials with strictly interlacing real roots.*

Proof. Equations (3) and (2) give a zero third-order coefficient minor whose corresponding principal minor is nonzero. This directly contradicts the asserted equivalence. □

3 Formal verification and its boundary

`Subresultant.lean` defines polynomials by ascending rational coefficients, and generates the Sylvester submatrix for arbitrary p, q, i by shifted coefficients. Its determinant is ordinary recursive Laplace expansion. The coefficient-minor and subresultant definitions implement the convention in Section 1 directly.

The file proves exact degrees and monicity; the two determinant values; the first subresultant formula; the displayed matrix from its generated entries; rational factorizations and exact rational root sets; strict root ordering; and the Bézout identity. It also verifies the two factorizations and (1) by finite coefficient-list multiplication, independently of evaluating x.

The final theorem is `conjecture_00000005613_false`. It negates `VanishingEquivalence`, the claimed equivalence restricted to monic degree-three/degree-two inputs satisfying the explicit factorization-based strict-interlacing condition. This is a restricted instance of the stated universal clause, not an assertion that every remaining undefined term has been formalized.

Scalar-domain boundary. Lean's scalar type here is `Rat`, not a formal type of real or complex numbers. The coefficient identities are checked exactly over $\mathbb{Q}$; their transfer along the usual embeddings into $\mathbb{R}$ and $\mathbb{C}$, and the resulting real-root and polynomial-coprimality interpretations, are the elementary mathematical steps explained above. The theorem `no_common_root` itself quantifies over rational inputs; it is not advertised as a formal general polynomial gcd theorem.

Strict compilation under Lean 4.31.0 succeeds with only `Std`. All theorem axiom closures use at most `propext`, `Classical.choice`, and `Quot.sound`; there are no custom axioms, admitted proofs, or native-decision shortcuts. The supplied README gives the strict build and kernel-replay commands. The separate Python script recomputes the generic matrix, determinants, coefficient products, and Bézout identity using exact rational arithmetic; it is supplementary, not a substitute for the Lean proof.

References

[1] Michael Kerber. *Geometric Algorithms for Algebraic Curves and Surfaces.* Dissertation, Universität des Saarlandes, 2009, Section 2.3, Definitions 2.3.10–2.3.12, p. 30. Author-hosted full text.